

in their device type specification.

9.2.7.3. No Cycles

- An endpoint SHALL NOT include itself in its PartsList attribute.
- Cycles where an endpoint is a (direct or indirect) ancestor of itself SHALL NOT occur.
- An endpoint (except endpoint 0) SHALL have a single parent, i.e. there SHALL be a single path from the Root Node to each endpoint.

9.2.8. Root Node Endpoint

A root node device type is a device type that is specified to be supported by all nodes in a fabric. Its main function is to describe the node and what endpoints the node supports.

The following requirements apply to root node endpoints:

- There SHALL be only one root node endpoint for the node.
- A root node device type SHALL be a [singleton](#) (representing the entire node) on the root node endpoint.
- The root node endpoint MAY contain additional [utility device types](#).
- The PartsList of the Descriptor cluster on the root node endpoint SHALL list all endpoints on the node, except the root node endpoint.
- The root node endpoint SHALL NOT exist in any other endpoint's PartsList on the node.
- The root node endpoint requirements are defined by a node scoped device type.

NOTE

The root node endpoint is currently under-specified for nodes which are not commissionable. As a result, the root node endpoint on non-commissionable nodes MAY be absent or otherwise limited in functionality. This specification gap is expected to be addressed in a future revision.

9.2.9. Disambiguation

Using the terms and composition patterns described above, some endpoints will have sibling endpoints with an overlap in application device type(s). For this case, the **Duplicate** condition has been defined (see the Base Device Type in the Device Library).

- A device type definition MAY define a method for disambiguation when the Duplicate condition applies to either the endpoint with that device type or to its children.
- If no disambiguation method is defined in the relevant device type definitions, an endpoint to which the Duplicate condition applies SHALL have a TagList attribute in its Descriptor cluster and its set of tags SHALL be distinct from the set of tags of any sibling endpoint with which it has an overlap in application device type(s). When comparing sets of tags, ordering of the list SHALL be ignored.
- The TagList feature and attribute MAY also be used in other cases to provide guidance to a client